



SYMBiosis SG: TURNING INDUSTRIAL WASTE INTO CERTIFIED PROFIT

The Hybrid Digital Platform for APAC Industrial Circularity

Our Mission is Dual-Focused:

- **Catalyzing Large-Scale Regeneration:** Creating self-sustaining, circular industrial systems across APAC.
- **Transforming Material Lifecycles:** Maximizing reuse of materials like plastic and reducing reliance on high-carbon virgin inputs.

THE TRIPLE CRISIS: WHY CIRCULARITY FAILS TODAY



THE FINANCIAL LOSS

Industrial companies treat valuable by-products as garbage, paying mandatory fees (e.g., S\$88/tonne for incineration). This creates a massive net financial loss for material the market actually needs.



THE TRUST & CONFIDENTIALITY GAP

Digital B2B waste exchanges fail due to **zero trust**. Sellers fear sharing proprietary data (chemical composition, volume) and buyers will not risk expensive production lines on unverified "waste."



THE CLIMATE TRAP

The lack of a trusted, verified local supply chain forces manufacturers to default to **100% high-carbon virgin materials**, preventing access to global sustainable incentives (e.g., UCO-to-SAF markets).

The current system is structurally built for linear, high-cost disposal, not circular, profitable reuse.

SYMBiosis SG: THE HYBRID DIGITAL BROKERAGE

We provide the secure, digitally-tracked transaction architecture that the Singapore industrial market—and the wider APAC region—has been missing.



The Human Broker: Building Trust

- **Role:** Acts as the essential, **vett**ed intermediary.
- **Compliance Action:** Broker verifies the Supplier's UEN/licensing and validates the material's **Certificate of Analysis (COA)** to guarantee integrity.
- **Outcome:** Guarantees the Seller receives the **maximum possible market value** for their asset.



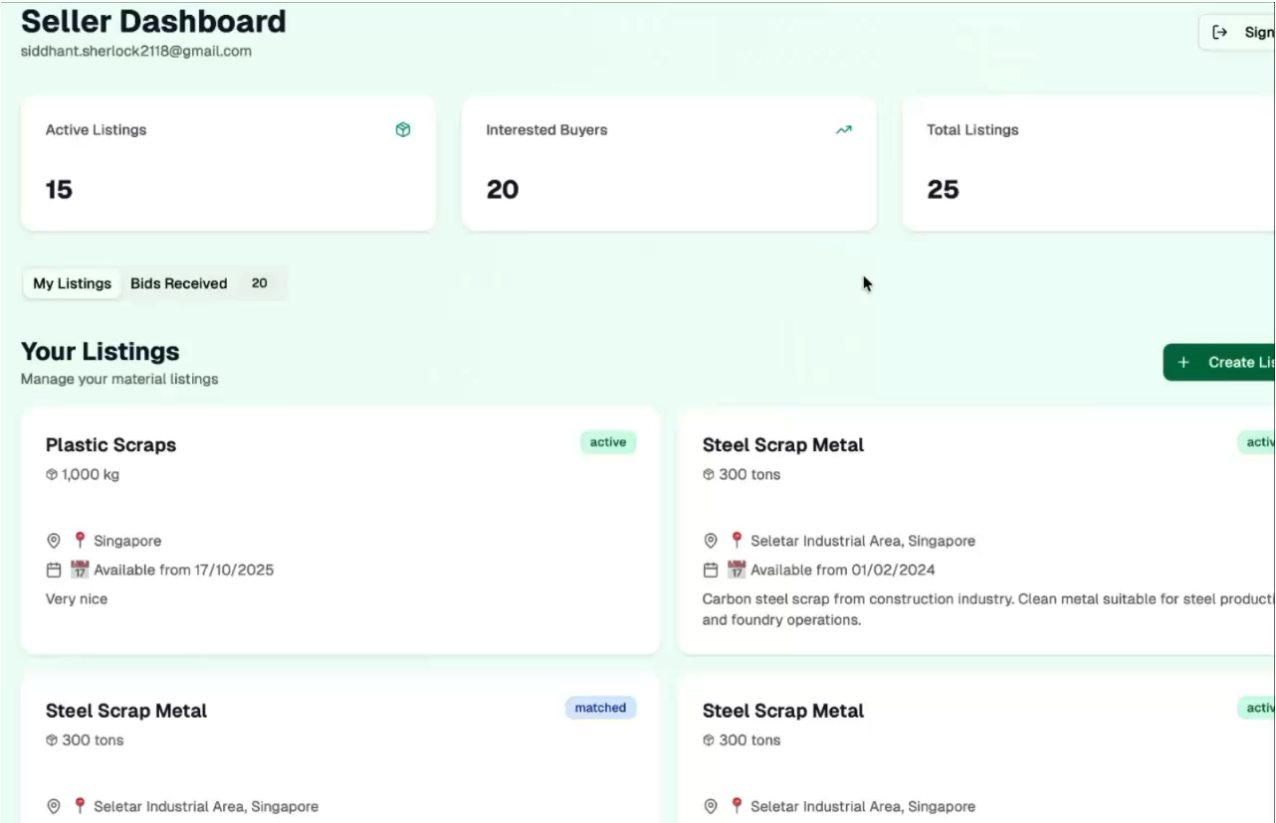
The Bidding Engine: Maximizing Value

Sellers list material, and Buyers submit bids. This guarantees the Seller receives the maximum possible market value for their asset, solving the "Fixed Price" and "Value Cap" problem.



The Digital Core: Enabling Scale

- **Resource Genome:** A proprietary database that matches material based on certified chemical specs.
- **Confidential Bidding Engine:** Buyers submit **confidential bids** over a set window.
- **Digital Traceability:** Provides an immutable Chain of Custody (CoC) record for all transfers.



VALUE CREATION & COMPETITIVE EDGE: De-Risking the Circular Economy

Question: "Why do they need a middleman?"

Answer: We sell certified quality, security, and market pricing power.

Feature	Direct Exchange (Status Quo)	Symbiosis SG (The Essential Broker)
Data Quality	No quality assurance. Based solely on Seller's word.	Provides Certified Quality (COA/digital vetting). Waste is now a trusted raw material.
Liability & Risk	Buyer exposed to full risk if material is contaminated (costly machinery downtime).	Transfers product liability to transacting parties, while providing auditable defense against fraud .
Pricing Mechanism	Fixed, low price based on disposal cost alternative.	Maximizes Seller Revenue via competitive, confidential bidding engine.



AUDIENCE RELATIONSHIP: WHY SELLERS NEED US MORE

The structural pressures of mandatory reporting and rising costs guarantee our initial supply—the hardest barrier in any marketplace development.

“

The Question

"Why do they need us?"

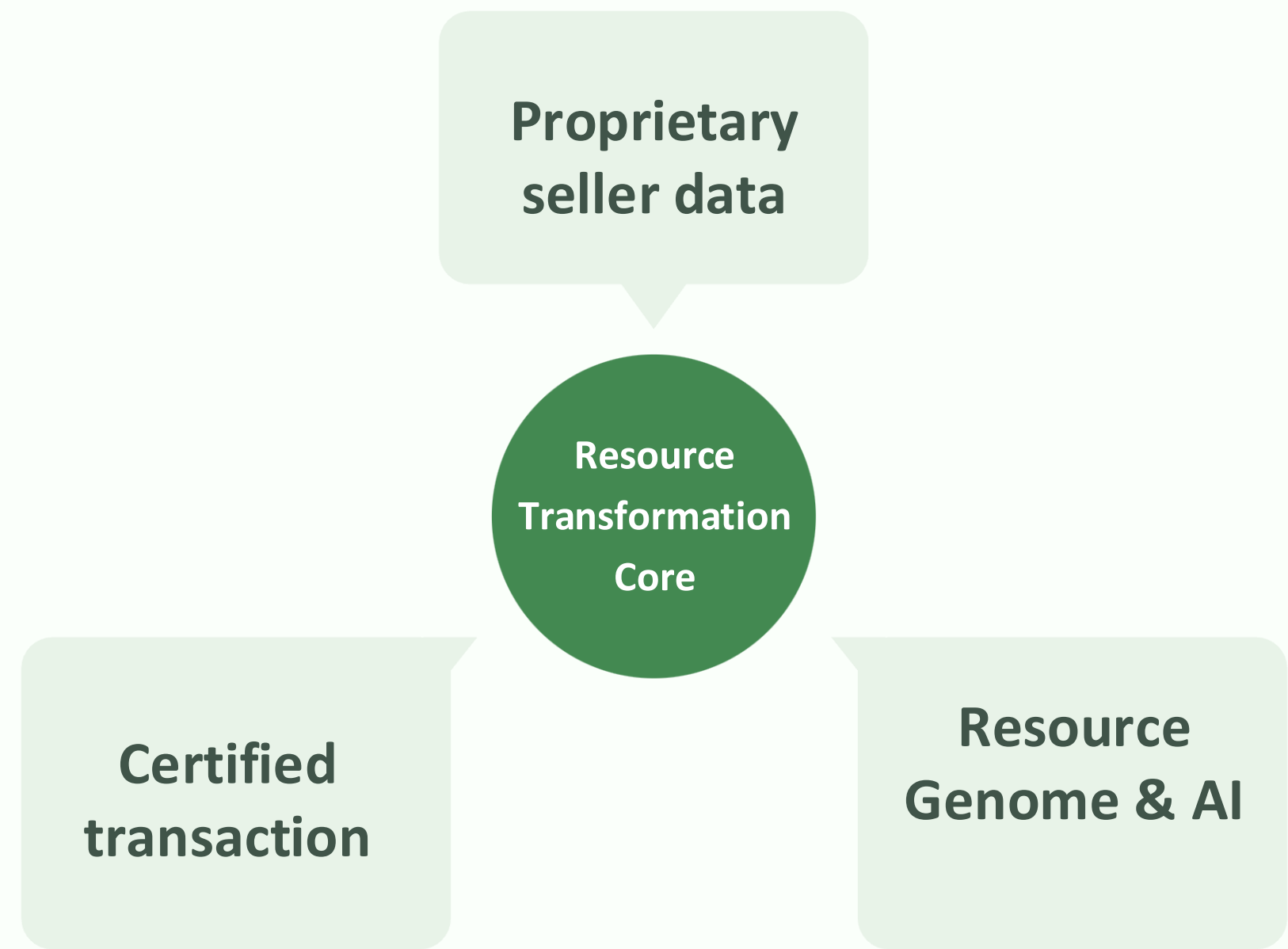
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The Answer: Sellers Need Us

The looming threat of mandatory **2025 sustainability reporting**, combined with rising disposal costs, forces large industrial players to find a trusted, certified disposal-to-revenue channel. This urgent need guarantees initial platform adoption.

PRODUCT ARCHITECTURE: OUR DIGITAL CORE



Our system transforms confidential industrial data into verified, tradable assets.

- **Resource Genome:** Database mapping specific chemical profiles to potential industrial applications.
- **AI Matching Engine:** Algorithms suggesting optimal buyers based on material specifications and production needs.
- **Verified Output:** Ensures all material leaving the system is certified and fully traceable.

Market Opportunity:

The total addressable market (TAM) for industrial recyclables in APAC is vast, driven by high-value materials like metals and plastics.

\$21B

Total Addressable Market (TAM)

APAC market for industrial recyclables (Metals, Plastics, Paper, E-waste, Textiles).

\$7B

Serviceable Available Market (SAM)

The portion of the TAM that can be serviced by a digital platform (31.6% of TAM).

\$35M

Serviceable Obtainable Market (SOM)

Realistic capture in the first 3-5 years (0.5% of SAM), focusing on high-priority materials.

Business model:

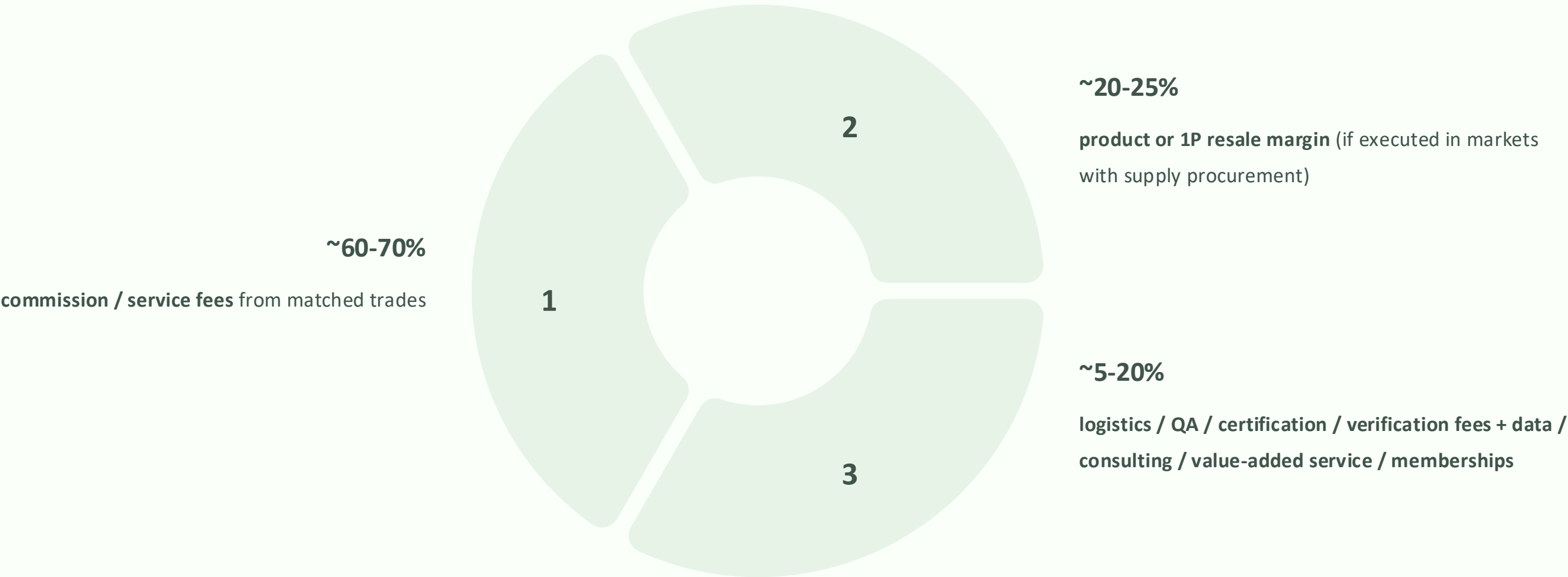
Model	Description	Revenue Mechanism	Example
3P (Marketplace)	Platform connects buyers & sellers directly.	Commission (e.g., 5–10%) per transaction.	WasteTrade (UK), Wastelink (India)
1P (First-Party Reseller)	Platform buys waste/surplus from one party → sells it onward (often in bulk).	Product resale margin (e.g., 20–25%).	Metaloop (Austria), ScrapMonster (China)
Hybrid	Does both — starts with 1P to ensure liquidity, transitions to 2P as network grows.	Both commission and resale margins.	ResourceCo (Australia), Circular.co (Denmark)

Business Strategy

Assuming we do not get Series A funding:

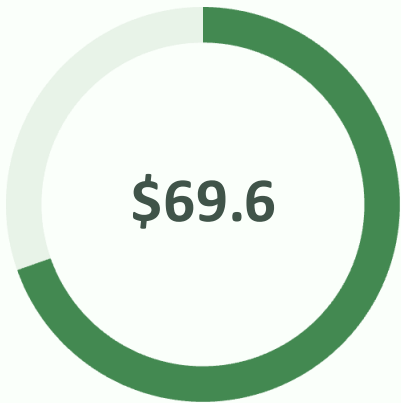
We use 1P resale initially to guarantee liquidity, generate higher margins, and build verified data on quality, traceability, and CO₂ savings. As supply and demand grow, we transition to a 3P marketplace, scaling efficiently while retaining control over critical streams.

Revenue mix



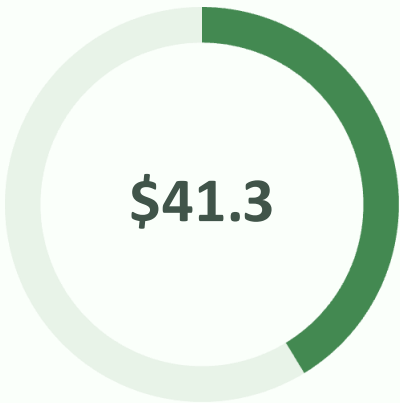
Circular Profit Per Tonne: Metrics That Matter

Our platform generates substantial financial and environmental ROI for both buyers and sellers on every tonne traded.



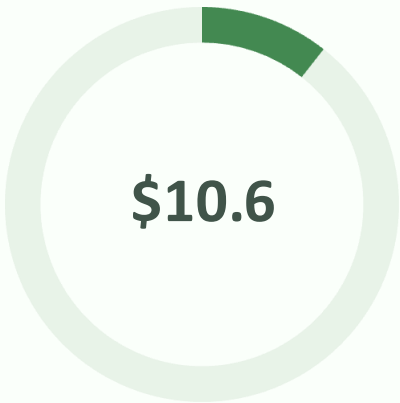
Avg Seller Gain Per 100kg

Increased profit vs. disposal cost.



Avg Buyer Saving Per 100kg

Cost reduction vs. virgin material procurement.



Platform Rev Per 100kg

Rev gained per tonne of material.



CO₂ Saved Per 100kg

Emissions avoided per tonne of material.

How individual companies can profit

Company size	Tonnes / yr	Seller total gain (USD/yr)	Buyer savings (USD/yr)	Platform revenue (USD/yr)	CO ₂ saved (t/yr)	Energy saved (MWh/yr)
Small	1,000	1,197,000	650,000	171,000	3,460	11,840
Mid	10,000	11,970,000	6,500,000	1,710,000	34,600	118,400
Large	100,000	119,700,000	65,000,000	17,100,000	346,000	1,184,000

Scaling Roadmap and Operational Metrics

Our phased approach focuses on building initial liquidity and trust before scaling across Southeast Asia (SEA).



Annex

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ANNEX 1: SG Climate Roadmap

Audit Defense and De-Risking

We provide the Certified Audit Trail required for external limited assurance (mandated from FY2029). Our digital records and COAs prove you acted responsibly.

Profit Maximization

Our Confidential Bidding Engine guarantees true market value. We eliminate the pricing risk and ensure you capture the maximum sustainable premium (profit) for your material.

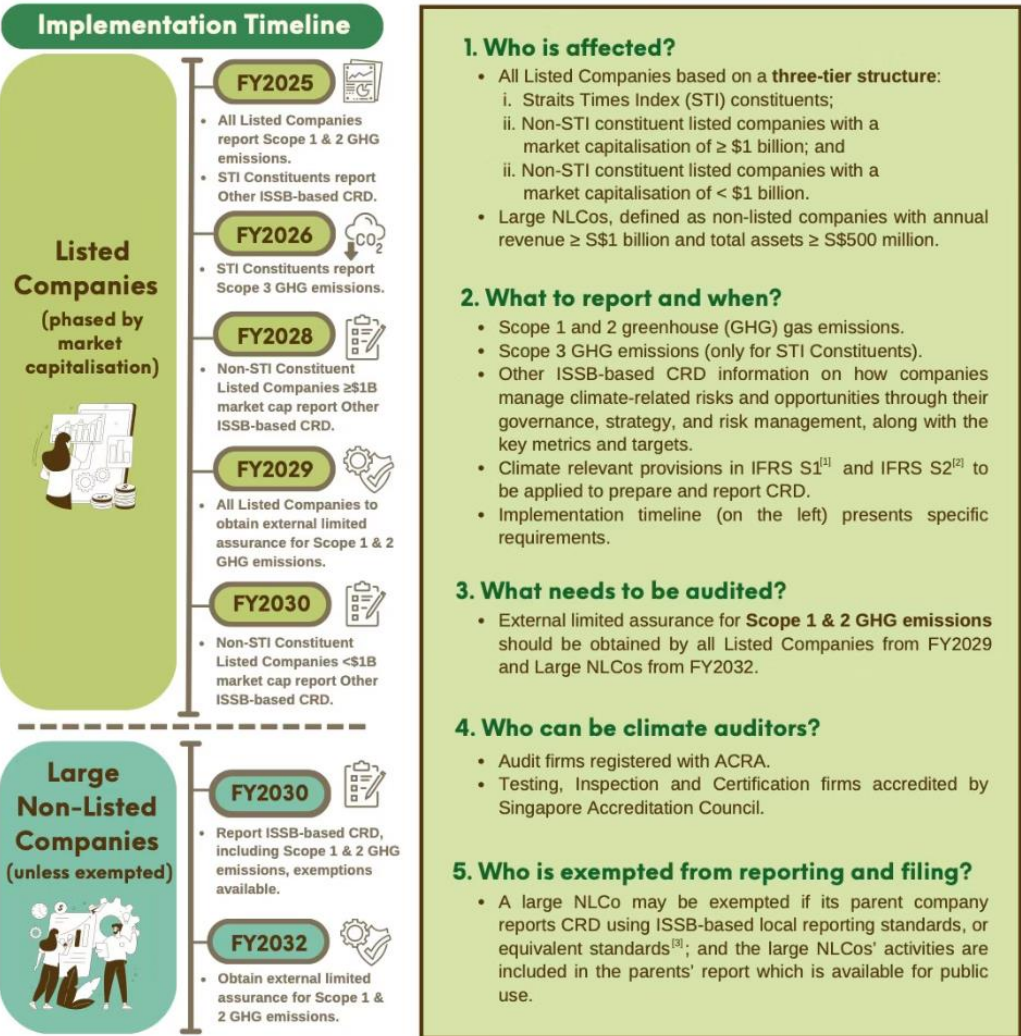
Compliance Automation

We make you a Preferred Supplier. We certify your material as a verified low-carbon input, ensuring your largest, most demanding clients can easily meet their Scope 3 obligations by buying from you.

The Roadmap mandates that waste must be **transparently tracked, audited, and reduced**. Symbiosis SG transforms this regulatory burden into a guaranteed profit center and a compliant sales channel, saving you audit failure, competitive risk, and the disposal fee.

Climate Reporting and Assurance Roadmap in Singapore

This roadmap aims to uphold Singapore's attractiveness as a global business hub while contributing to the national agenda on sustainable development under the Singapore Green Plan 2030. As the global momentum for bolder climate action intensifies, companies face increasing scrutiny to decarbonise and transition towards more sustainable practices. Those who are able to demonstrate through climate-related disclosures (CRD) based on internationally recognised standards, that they are ahead in their decarbonisation journeys, stand to benefit from access to new markets, customers, and financing.



[1] IFRS S1 General Requirements for Disclosure of Sustainability-related Financial Information sets out how to prepare the content and report sustainability-related financial disclosures.

[2] IFRS S2 Climate-related Disclosures require an entity to disclose information about its climate-related risks and opportunities that is useful to users of general-purpose financial reports in making decisions relating to providing resources to the entity.

[3] As international sustainability disclosure standards continue to evolve, ACRA will issue guidance on the standards deemed equivalent to guide implementation.

ANNEX 2: License

1. The Key Distinction: Brokerage vs. Facility Operations

The NEA's GWDF regulations are designed to regulate the **physical handling** of waste. Our platform does not meet the definition of a GWDF.

NEA GWDF Definition

"Facility that receives, stores, processes, or treats general waste."



General Waste Disposal Facility

The National Environment Agency



Symbiosis SG's Operational Status

Symbiosis SG is a Software Platform. We are a digital matchmaking service and broker of information. We do not own, touch, or store any physical material. All waste remains on the **Seller's premises** until direct transfer to the **Buyer's premises**.

Conclusion

We are Exempt from GWDF Licensing.

Our low-overhead, digital model allows us to scale instantly without the multi-million dollar capital expenditure, permitting, and inspection required for a physical facility.

2. Explanation: Compliance as a Service Layer

Instead of holding the license, we enforce compliance among the parties, turning regulatory necessity into our business value

Compliance Requirement	How Symbiosis SG Manages Risk and Responsibility
Buyer Licensing (GWDF)	We mandate that all Buyers seeking to receive and process industrial by-products must submit their valid GWDF or relevant industrial processing license during UEN verification. We do not create the regulatory burden; we verify that the Buyer is legitimate and capable of safe processing.
Data & Traceability	We provide the necessary digital tools to help the Seller comply with NEA's documentation requirements (e.g., logging source, type, and quantity). Our Digital Chain-of-Custody (CoC) provides the immutable record required for external assurance.
Liability Mitigation	All transactions are covered by contractual terms. We legally position ourselves as the broker of information, transferring all product liability (quality/safety) directly between the Seller and the Buyer.

ANNEX 3: Bidding System

Bidding System for Industrial Waste Trading

Advantages & Considerations

Market-Driven Pricing

Enables sellers to achieve optimal value through competitive market dynamics and transparent price discovery mechanisms.

Fosters Competition

Encourages multiple buyers to participate actively, driving innovation and efficiency in waste material transactions.

Demand Intelligence

Provides valuable insights into market demand patterns, helping stakeholders make informed strategic decisions.

Collusion Risk

Potential for bid manipulation or coordinated behavior among buyers that undermines fair market competition.

Monitoring Complexity

Requires sophisticated oversight systems and substantial resources to detect irregularities and enforce compliance.

Trust Infrastructure

Demands robust verification mechanisms and transparent processes to build participant confidence and system integrity.

ANNEX 4: Competitor Analysis

Extended Competitor Analysis: Industrial Waste & Resource Exchange Platforms

Symbiosis SG uniquely differentiates itself through transparent bidding, verified listings, and APAC-focused localization, addressing critical gaps in trust, pricing, and scalability where existing platforms struggle.

Platform / Initiative	Region	Model Type	Revenue Focus	Strengths	Weaknesses	✅ Symbiosis SG Advantage
Metaloop	Austria (EU)	Hybrid platform + SaaS	20–25% margin + commission	Deep metal recycling network; strong EU reputation; SaaS evolution	Metal-only focus; legacy systems; opaque pricing structures	Transparent bidding enables fairer, data-driven price discovery across all material types
WasteTrade	UK / APAC	2P B2B marketplace	6–10% transaction fee + data subscription	Simple UI; diverse material range; rapid geographic expansion	Weak verification processes; inconsistent pricing; limited buyer trust	Verified listings and structured bidding prevent price volatility and build credibility
ResourceCo	Australia	1P physical processor	~20% material margin	Extensive processing infrastructure; established market reputation	Asset-heavy model; geographically constrained operations	Platform-based approach achieves similar outcomes without capital-intensive assets
Circular.co	Denmark / EU	SaaS marketplace + analytics	Subscription + data monetization	Integrated ESG reporting and sustainability metrics	Manual pricing processes; poor ASEAN market penetration	Automated pricing through live bidding system and demand-side analytics
ScrapMonster / ScrapChina	China	1P reseller + lead generation	~25% resale spread	High transaction volume; established metal sector presence	Static pricing; no competitive bidding; upstream focus only	Open bidding increases downstream participation and market efficiency
NISP	UK	Human-facilitated matching	Government funding	Proven circular collaboration track record; trusted model	Non-scalable; relies on manual negotiation processes	Digitizes matchmaking with automated bidding for scalability and speed

The Industrial Waste Exchange Landscape: Persistent Gaps

- Opaque Pricing & Inconsistent Verification
- Limited Scalability & Geographic Constraints
- Material-Specific Focus

Symbiosis SG: Bridging the Gaps

- Transparent Bidding
- Verified Listings & Participants
- Scalable Digital Matchmaking
- APAC Localization

ANNEX 5: Tradable Material

“What if a company sells unlicensed or hazardous material through your platform?”

Our strategy is to use our digital vetting process to ensure the platform only handles non-hazardous industrial by-products being transferred to licensed processors.

1. The Material Exclusion Zone: What We Matchmake

Symbiosis SG strictly define its scope to avoid the regulatory complexity of hazardous waste (which requires separate Toxic Industrial Waste Collector licenses, specialized handling, etc.).

Material Category	Examples	Symbiosis SG Policy	Rationale
Ideal for Symbiosis (Non-Hazardous By-products)	Clean Plastic Off-Cuts (HDPE, PP), Cooking Oil (UCO), Spent Coffee Grounds, Wood Waste (untreated), Food Processing Residue.	Full Vetting & Matching. These are suitable for direct material substitution.	Low Risk: These materials are not legally classified as "toxic" and require general waste licenses or standard industrial licenses for handling/processing.
Material to Flag and Avoid (Hazardous/Toxic Waste)	Spent Solvents, Heavy Metal Sludge, Chemical Residues, Spent Acids/Alkalis.	Strictly Prohibited. Our platform immediately flags and rejects listings that match known hazardous chemical profiles.	High Risk: Trading these requires specialized, dedicated licenses, high insurance, and specialized transport that exceeds our startup scope.

ANNEX 6 : Verification (Trust)

Verification of the Material (The Data Layer)

We do not test the material; we verify the integrity of the data about the material, providing the necessary audit trail for external assurance (mandated from FY2029).

COA (Certificate of Analysis) Mandate

Sellers provide internal Quality Control (QC) data logs and specifications (FFA%, MFI). Our Broker validates these logs against industry standards, creating a mandatory Digital COA Specification for the listing.

Digital Chain-of-Custody (CoC)

Every material lot is tagged with a digital ID. When the material is picked up/delivered, the Buyer and Seller verify the transfer digitally, creating an immutable, location-specific record of the transaction.

Proof of Integrity

The Buyer's and Seller's UENs are logged with the material's specifications. This verifiable provenance eliminates the risk of fraud (like mixing in cheaper, non-sustainable virgin materials).

✔ **Why it's Essential:** For the Buyer, CoC proves that the low-carbon, verified UCO they bid on is the exact material they received, and that no high-carbon virgin oil was illegally mixed in during transit. This is the foundation of the Audit Trail for their ESG report.

Verification of the Company (The Trust Layer)

The platform focuses on verifying the integrity of the businesses involved, making them Certified Sustainable Transactors.

Mandatory UEN Vetting

We verify the legal standing and licensing of every company (Buyer and Seller) during onboarding to ensure legitimacy and compliance with NEA/GWDF rules.

Performance Certification

Instead of a traditional "feedback score," the platform tracks successful, high-value closed transactions. This builds a publicly referenceable track record (e.g., "Company X diverted 10,000 MT of plastic via Symbiosis SG").

Annex 7: Material Pricing

Per Tonne economics for different materials

Material	Virgin price (USD/t)	Sale price (USD/t)	Disposal avoided (USD/t)	Seller total gain (USD/t)	Buyer saving USD/t	Platform Rev@ 6% Commission USD/t	Platform Rev@ 14.3% Blended Take USD/t
Steel (HMS)	\$814/t (Hot-rolled coil index, TradingEconomics snapshot Oct 10, 2025).	\$320/t (HMS scrap modelling price recommended)	40	494	565	19.2	45.76
Aluminium	\$2,749 – \$2,750/t (LME aluminium price ~ Oct 2025).	\$2,200/t (scrap aluminium modelling price).	40	550	400	132.0	314.60
Paper (OCC)	~\$900 / t (virgin pulp/packaging board regional values; IMARC / Fastmarkets data)	\$220/t (OCC modelling price).	60	680	380	13.2	31.46
Plastics (rPET proxy)	~\$850 – \$1,090 / t (PET resin indices: IMARC reported PET prices ~ \$0.85–1.09/kg = \$850–1,090/t depending on region).	\$1,050/t (rPET modelling price used earlier)	75	40 (PET pricing often flips: sometimes virgin cheaper, sometimes rPET trades at a premium due to shortages or quality.)	300	63.0	150.15
E-waste (mixed)	Really Depends	\$450/t (mixed feedstock proxy used earlier)	300	Varies a lot	200	27.0	64.35

How the 14.3% take rate was derived

That blended figure came from this assumed mix:

1P share of GMV = 20%; 1P gross margin = 20% → contributes $0.20 \times 0.20 = 4.0\%$ of GMV.

3P share of GMV = 80%; commission on 3P = 6% → contributes $0.80 \times 0.06 = 4.8\%$ of GMV.

Services / QA / logistics fees = 5.0% of GMV.

Data / subscriptions (annualized) = 0.5% of GMV. Total blended take = $4.0 + 4.8 + 5.0 + 0.5 = 14.3\%$.

Annex 8: Case Study

Mixed-average per-tonne (our composition: 35% steel, 15% Alu, 20% paper, 20% plastics, 10% e-waste - mixed)

This gives a single-per-tonne number you can multiply by any annual tonnage.

Total Weighted Seller Gain ≈ 696.1 USD/t

Total Weighted Buyer Saving ≈ 413.75 USD/t

Our platform

- 44.4 USD/T (purely from commissions)
- 106 USD/T (combined)

Company size	Typical annual throughput (t/yr)	Notes
Small	500–2,000 t/yr	Usually local scrap yards or small industrial generators; may handle single material streams like OCC or small plastics volumes.
Mid	5,000–20,000 t/yr	Regional recyclers, industrial hubs, or medium-sized brokers; multiple material streams.
Large	50,000–200,000 t/yr	National recyclers or integrated 1P/2P platforms; handles metals, plastics, paper, e-waste at scale.

Annex 9

Any other questions:

Bidding works better for **homogeneous, high-volume commodities** with established specs (e.g., financial securities, standard metals like copper cathode). In fragmented industrial waste streams, **direct negotiation + 1P guarantee + optional analytics** is more practical and less risky.

1. Supplier Pickup / Collection

The platform either coordinates pickups via its own vehicles or partners with local logistics providers.

They ensure timely collection, proper handling, and sometimes initial quality checks at the supplier site.

2.Transport to Storage / Sorting Facility

Materials are transported to a central warehouse, yard, or recovery facility.

At this stage, they may perform:

Sorting by type or grade (e.g., separating metals, plastics)

Weighing and documentation

Initial quality testing (e.g., purity, contamination levels)

3.Storage / Temporary Holding

Materials are often stored until buyer orders are ready.

Storage ensures the platform can guarantee supply even if sellers are irregular.

4.Resale & Delivery to Buyers

Once a buyer is matched:

Materials are packaged/loaded according to buyer requirements.

Transport can be handled by in-house fleet or third-party carriers, sometimes integrated via logistics APIs.

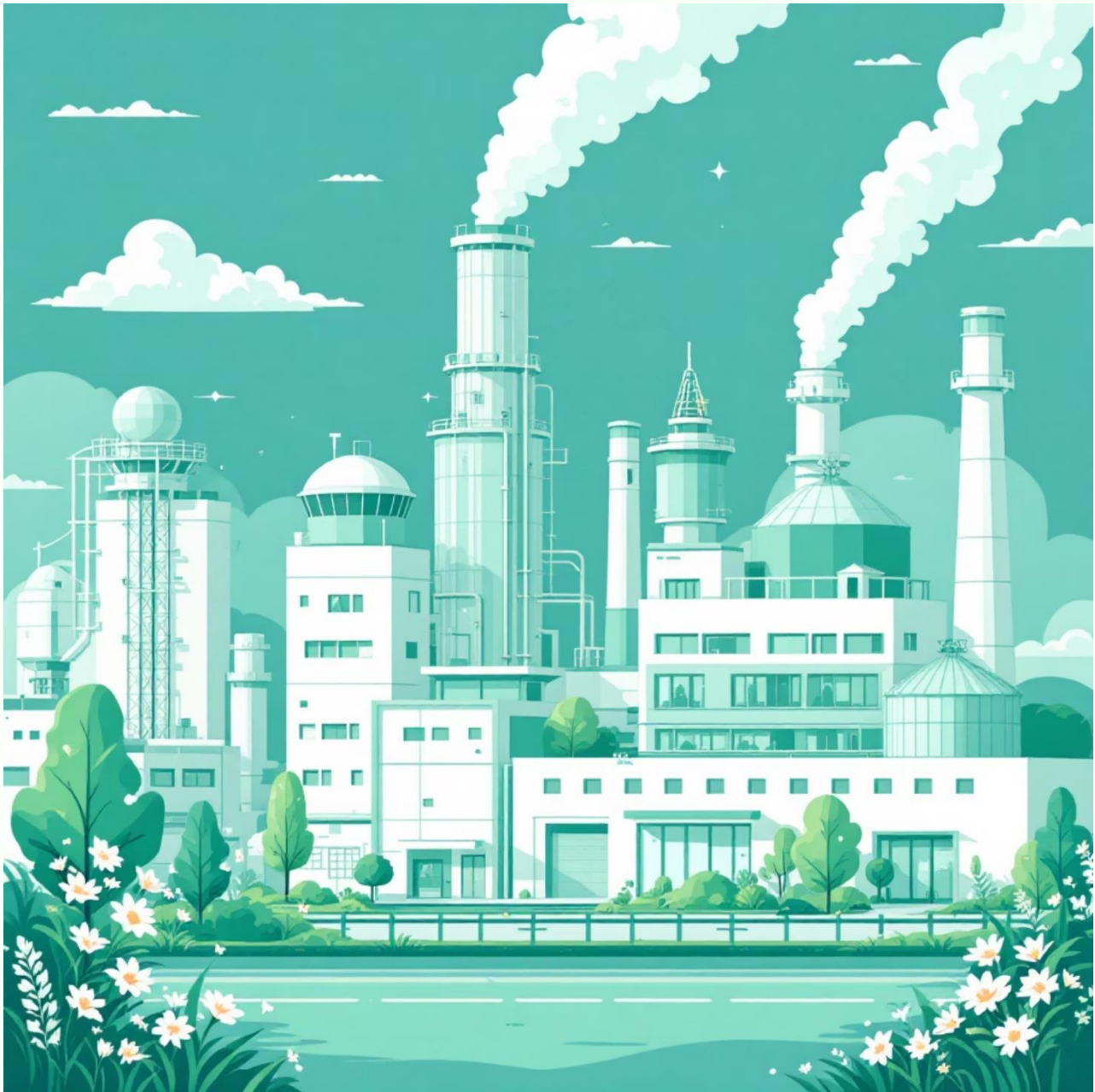
Traceability and CO₂ reporting are included as part of delivery documentation.

Annex 10: USP

Unique Selling Point

Transforming Industrial Waste Trading

We turn industrial waste trading from a fragmented, risky, and resource-intensive process into a **streamlined, value-maximizing system**—delivering not just convenience, but efficiency, risk reduction, and revenue optimization.



ANNEX 11: Buyer POV

Real-World Case Study: Wood Waste to Pallets in Singapore

The best example involves an industrial by-product that used to be pure waste, but is now a circular commodity: **Industrial Wood Scraps and Sawdust**.

The Companies Involved (The Symbiosis)

- The Seller (Generator): Timber, Furniture, and Crating Companies** in industrial estates (like Sungei Kadut). They generate tonnes of waste sawdust, off-cuts, and broken pallets.
- The Buyer (Processor): LHT Holdings Ltd, ChopValue, etc..**(a real Singaporean company). They specialized in converting wood waste into environmentally friendly products like **recycled wood pallets (EPALs)** and wooden components.

The Situation: Before Symbiosis SG

Metric	Seller's Situation (Timber Company)	Buyer's Situation (LHT Holdings)
Material Status	100% Waste. Had to pay S\$150 - S\$250/tonne(estimate) to transport the bulky wood waste to an incinerator or landfill.	Had to constantly buy virgin timber (high cost, high carbon footprint) to manufacture new components, creating supply chain vulnerability.
Financial Outcome	Net Loss. Paid a disposal fee.	High Input Cost. Paid premium prices for virgin wood.
Environmental Outcome	Polluting: CO2 emissions from incineration.	Linear: Perpetuated resource depletion and deforestation.

Annex 12: Buyers

Identifying Primary Buyers for Symbiosis SG

Your platform's value is providing a **vetted, bulk supply** of feedstock. This means your ideal buyers are those whose entire business model relies on turning waste into a resource (The Processor) or using certified recycled content.

1. Buyers for Wood By-Products (Off-cuts, Sawdust, Pallet Scraps)

This material is valuable because it is generally clean and uniform.

Buyer Company	Material Use	Why They Need Symbiosis SG
LHT Holdings	Converts wood waste into "Technical Wood" (engineered compressed wood) for pallets and crates.	They are the ideal bulk Buyer. They need a continuous, high-volume supply of clean wood fiber to sustain their factory operations..
Bluefield Renewable Energy	Produces certified biochar for green concrete and soil amendments.	They need consistent, segregated wood biomass to feed their pyrolysis units. Our platform provides the audited data (type, volume) to meet their complex process requirements.
RExus Bioenergy / ESGL	Converts waste wood into renewable electricity (Waste-Wood-to-Energy).	They require high-volume, reliable feedstock to keep their plants running efficiently. Our platform acts as the digital procurement channel for their fuel supply.
ChopValue Singapore	Repurposes used chopsticks into premium products.	While niche, they represent the small-to-medium enterprise (SME) who needs a specific, clean waste stream. They are high-margin Buyers who value the ease of supply what we offer.

ANNEX 13: Other buyers

Buyers for Non-Wood By-Products (UCO and Plastic Off-Cuts)

While not listed, the presence of major global energy players in Singapore confirms the market for these products:

- **Used Cooking Oil (UCO) Buyers:** Companies like **Neste** (which operates a massive biorefinery in Singapore) are ideal targets. They are desperate for certified, low-carbon feedstocks like UCO to make Sustainable Aviation Fuel (SAF). They need the Chain-of-Custody your platform provides to prove their SAF is sustainable for global export.
- **Plastic Off-Cut Buyers:** Companies that run pelletizing operations to sell recycled plastic resin would buy high volumes. These are often unnamed B2B chemical firms or plastic product manufacturers.